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Options and Futures

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Stochastic Processes and Itô Calculus

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- Stochastic processes
- Brownian motion and Itô processes
- Itô's lemma
- Stock prices as geometric Brownian motions

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Stochastic Processes

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- Let $\mathcal{T}$ denote a discrete set of dates $\{0, 1, 2, \dots\}$ or a time interval, e.g. $[0, T]$ or $[0, \infty)$.
- Let Ω denote a set of states of the world. An element ω of Ω describes the state of the world from the beginning to the end of time $\mathcal{T}$.
- A stochastic process X is formally defined as a mapping from $\Omega \times \mathcal{T}$ into $\mathbb{R}$. We write $X : \Omega \times \mathcal{T} \to \mathbb{R}$; $X : (\omega, t) \mapsto X(\omega, t)$.
- For given time $t \in \mathcal{T}$, $X(\cdot, t) : \omega \mapsto X(\omega, t)$ is a random variable. We usually drop reference to ω and write $X(\omega, t)$ as $X(t)$ or X_t.
- For given $\omega \in \Omega$, $X(\omega, \cdot) : t \mapsto X(\omega, t)$ is just a function of time, referred to as a sample path of X.
- When $\mathcal{T}$ is a continuum, we say X is a continuous-time process (vs. discrete-time process).

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Standard Brownian motion

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A standard Brownian motion (or Wiener process) W is a continuous-time stochastic process which satisfies the following properties:

(i) $W_0 = 0$.

(ii) Normally distributed increments: For any time t and $s > t$,

$$W_s - W_t \sim \mathcal{N}(0, s - t).$$

(iii) Independent increments: For any times $t_0, \ldots, t_n$ such that $t_0 < t_1 < \ldots < t_n$, the random variables $W_{t_0}, W_{t_1} - W_{t_0}, \ldots, W_{t_n} - W_{t_{n-1}}$ are independent.

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Simulating a standard Brownian motion over $[0, T]$

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- Consider n subintervals of size $\Delta t := \frac{T}{n}$ and let $t_i = i\Delta t$, $0 \le i \le n$.
- Note that a standard Brownian increment over any subinterval is normally distributed with mean 0 and variance Δt, i.e. it has the same distribution as $z\sqrt{\Delta t}$, where z is a standard normal random variable.
- For given n, let $\{z_i\}_{i=1}^n$ denote a sequence of independent standard normal variables and define recursively

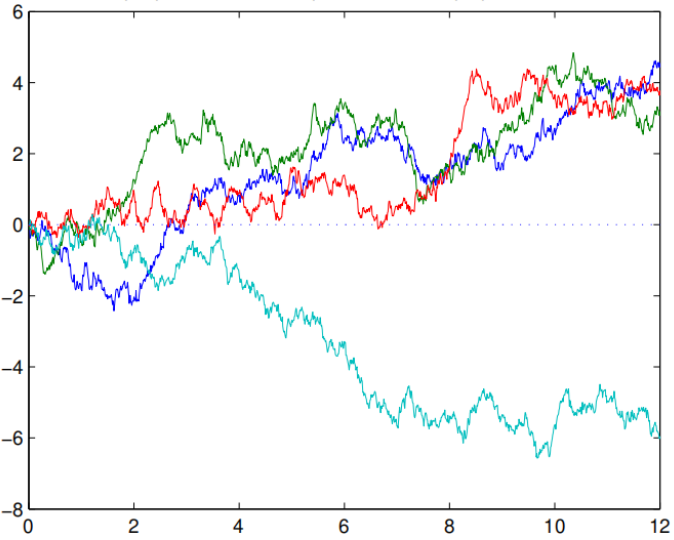
$$W_0 = 0,$$
$$W_{t_{i+1}} = W_{t_i} + z_{i+1}\sqrt{\Delta t}, \quad \text{for } i = 0, \ldots, n-1.$$

- A continuous process W obtains by linear interpolation over intervals $[t_i, t_{i+1}]$.

Simulation example: Standard Brownian sample paths

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Sample paths of a Wiener process - 50 steps per unit of time



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Basic properties of standard Brownian motion

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- W has normal increments $\Rightarrow$ its distribution at all times is normal.

$$W_t = W_t - W_0 \sim \mathcal{N}(0, t).$$

- Let $s > t$. We can write $W_s = W_t + (W_s - W_t)$. Therefore, conditional on W_t, W_s is normally distributed, with

$$\mathbb{E}_t[W_s] = W_t$$

$$\mathrm{Var}_t[W_s] = s - t.$$

- In particular, W is a martingale [1].

[1] A stochastic process X is a *martingale* if at any point of time the conditional expectation of its future value is equal to its current value, i.e., $\mathbb{E}_t(X_s) = X_t$ for $t \leq s$.

(Generalized) Brownian motion

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- A (generalized) Brownian motion X with initial value X_0 is of the form

$$X_t = X_0 + at + bW_t,$$

  where a and b are constants and W is a standard Brownian motion.
- This process is also referred to as a Brownian motion with drift, or arithmetic Brownian motion, or simply a Brownian motion. Its evolution can be decomposed as a deterministic trend plus some Brownian noise.
- Note that $X_t \sim \mathcal{N}(X_0 + at, b^2 t)$.
- a is called the drift of X (it is the average change in X per unit of time), while b is called the diffusion or (instantaneous) volatility of X (it is the standard deviation per unit of time).
- The evolution of X can also be written as follows:

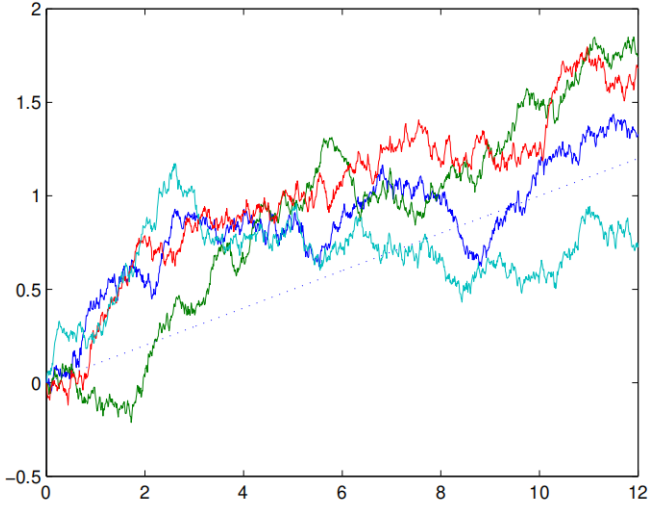
$$dX_t = adt + bdW_t.$$

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Sample paths of a Brownian motion with drift

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Sample paths of a generalized Wiener process for $a = 0.1$ and $b = 0.2$.



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- Let $s > t$. Then,

$$X_s - X_t = a(s-t) + b(W_s - W_t)$$

$$\sim \mathcal{N}(a(s-t), b^2(s-t)).$$

- Moreover, we can write $X_s = X_t + (X_s - X_t)$. Therefore, conditional on X_t, X_s is normally distributed, with

$$\mathbb{E}_t[X_s] = X_t + a(s-t)$$

$$\text{Var}_t[X_s] = b^2(s-t).$$

- A Brownian motion can be negative, which is problematic if we use it as a model for a stock price.

Itô processes

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- X is an Itô process if its random evolution is governed by

$$dX_t = a_t dt + b_t dW_t.$$

- a_t and b_t are allowed to be stochastic processes themselves, or deterministic functions of time.
  - a_t is called the drift process and b_t the diffusion process.
- A Brownian motion is thus a simple case of an Itô process.
- The above equation is also called a stochastic differential equation (SDE).
- Example:

$$dX_t = \mu X_t dt + \sigma X_t dW_t,$$

where μ and σ are constants. This process is called a geometric Brownian motion (GBM). In this context, μ is called the drift and σ the diffusion or (instantaneous) volatility of the GBM.

Functions of Itô processes

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- Suppose X follows an Itô process:

$$dX_t = a_t dt + b_t dW_t.$$

- Consider a twice continuously differentiable function

$$f : \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R},$$

$$f : (x, t) \mapsto f(x, t).$$

- Let Y be the process given by $Y_t = f(X_t, t)$.
- Itô's lemma tells us that Y is also an Itô process and allows us to compute its drift and diffusion using Itô's multiplication rules:

$$(dt)^2 = 0,$$

$$(dt)(dW_t) = 0,$$

$$(dW_t)^2 = dt.$$

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Itô's lemma

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- Itô's lemma says that the process Y given by $Y_t = f(X_t, t)$ is an Itô process, and

$$dY_t = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(dX_t)^2,$$

where, using Itô's multiplication rules,

$$dX_t = a_t dt + b_t dW_t,$$

$$(dX_t)^2 = a_t^2(dt)^2 + b_t^2(dW_t)^2 + 2a_t b_t(dt)(dW_t) = b_t^2 dt,$$

- This gives us:

$$dY_t = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial x}a_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}b_t^2\right)dt + \frac{\partial f}{\partial x}b_t dW_t.$$

Itô's lemma: intuition

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- Itô's lemma:

$$dY_t = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(dX_t)^2.$$

- The presence of the second-order term is the specificity of Itô calculus. In ordinary calculus, we would just write

$$dY_t = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX_t.$$

- But here $(dX_t)^2 = b_t^2(dW_t)^2$, which is not zero since $(dW_t)^2 = dt$.
- If W_t were a deterministic function of t, the rules of ordinary calculus would apply, and $(dW_t)^2$ would be equal to zero.
- Here, however, we have $\Delta W_t \sim \mathcal{N}(0, \Delta t)$, so that

$$\mathbb{E}[(\Delta W_t)^2] = \mathrm{Var}(\Delta W_t) = \Delta t.$$

- Heuristically, we can think of $(dW_t)^2$ as the limit of $\mathbb{E}[(\Delta W_t)^2]$ as the time interval becomes small.

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Example

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- Let $dX_t = adt + bdW_t$ and $Y_t = e^{X_t}$.
- In this case, $f(x,t) = e^x$, $\frac{\partial f}{\partial t} = 0$, $\frac{\partial f}{\partial x} = e^x$, $\frac{\partial^2 f}{\partial x^2} = e^x$.
- Therefore

$$\begin{aligned} dY_t &= e^{X_t} dX_t + \frac{1}{2} e^{X_t} (dX_t)^2 \\ &= e^{X_t}(adt + bdW_t) + \frac{1}{2} e^{X_t}(adt + bdW_t)^2 \\ &= e^{X_t}(adt + bdW_t) + \frac{1}{2} e^{X_t} b^2 dt \\ &= (a + \frac{b^2}{2}) e^{X_t} dt + b e^{X_t} dW_t \\ &= (a + \frac{b^2}{2}) Y_t dt + b Y_t dW_t. \end{aligned}$$

- Thus the exponential of a Brownian motion with drift and diffusion coefficients a and b is a geometric Brownian motion with drift and diffusion coefficients $a + \frac{b^2}{2}$ and b.

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Stock prices as geometric Brownian motions

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- In his PhD thesis "La Théorie de la Speculation" (1900), Louis Bachelier had a path-breaking idea – to represent the stochastic evolution of stock prices as a Brownian motion.
- As a matter of fact, the stochastic evolution of stock prices is better represented by a geometric Brownian motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

for some parameters μ and σ. Alternatively, we often write:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t.$$

- μ is called the drift or instantaneous expected return of the stock.
- σ is called the diffusion or (instantaneous) volatility of the stock.

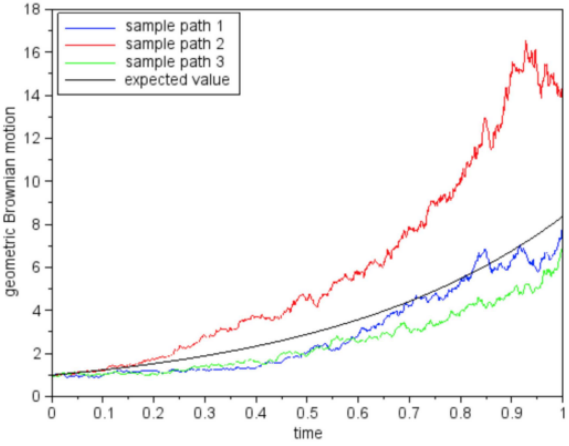
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- Suppose
$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$
- Consider $Y_t = \log S_t$. Applying Itô's lemma, we get
$$dY_t = \frac{1}{S_t} dS_t + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)(dS_t)^2$$
$$= \frac{1}{S_t}[\mu S_t dt + \sigma S_t dW_t] - \frac{1}{2}\frac{1}{S_t^2}\sigma^2 S_t^2 dt$$
$$= \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW_t,$$
which is a Brownian motion.
- This should not be surprising, given that we have already shown that the exponential of a Brownian motion is a geometric Brownian motion.

GBM sample paths

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Lognormal property of GBM

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- If a stock price S follows a GBM, then

$$d \log S_s = \left(\mu - \frac{\sigma^2}{2}\right) ds + \sigma dW_s.$$

- Integrating from t to T, we get

$$\log S_T - \log S_t = \int_t^T \left(\mu - \frac{\sigma^2}{2}\right) ds + \int_t^T \sigma dW_s$$

$$= \left(\mu - \frac{\sigma^2}{2}\right)(T-t) + \sigma(W_T - W_t).$$

- Hence the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \frac{\sigma^2}{2}\right)(T-t), \sigma^2(T-t)\right].$$

- Since $\log S_T$ is normally distributed, S_T is said to be lognormally distributed.

Expected stock price

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- We have shown that

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \frac{\sigma^2}{2}\right)(T-t), \sigma^2(T-t)\right].$$

- **Result:** If X is a normally distributed random variable, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\mathrm{Var}(X)\right].$$

- Therefore,

$$\begin{aligned}\mathbb{E}_t(S_T) &= \mathbb{E}_t\left[\exp(\log S_T)\right] \\ &= \exp\left[\log S_t + \left(\mu - \frac{\sigma^2}{2}\right)(T-t) + \frac{\sigma^2(T-t)}{2}\right] \\ &= e^{\log S_t} \cdot e^{\mu(T-t)} \\ &= S_t e^{\mu(T-t)},\end{aligned}$$

i.e., in expectation the stock price grows at rate μ.

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- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

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- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

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- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

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Thus we have:

$$\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}$$

Breeden-Litzenberger

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- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

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Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

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▷ Appendix B

- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

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USEFUL FACTS

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
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- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$

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